

Muhammad Aliyan



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Education

National University of Sciences and Technology (NUST), CEME

Rawalpindi, Pakistan

Bachelor of Engineering in Computer Engineering

Graduating May 2026

Experience

Full Stack Engineer

Nov 2025 – Present

Zapply (USA) – Remote

Full-Time

- Developing and maintaining scalable MERN applications for production environments.

Full Stack Developer

Aug 2025 – Nov 2025

3Dim (China) – Remote

Full-Time

- Built and deployed full-stack features using React.js, Node.js, and MongoDB.

Freelance Full Stack Developer

Jun 2023 – Present

Fiverr – Remote

Self-Employed

- Delivered end-to-end web solutions for international clients, including dashboards and APIs.

Full Stack Developer Intern

Jun 2025 – Aug 2025

Zapply (USA) – Remote

AI/ML Intern

May 2025 – Jul 2025

CARE

Final Year Project

Secure Communication Framework for Embedded RISC-V Microprocessors

May 2026

Verilog, FPGA, Vivado, Python

- Designed and implemented a hardware AES-ECB encryption coprocessor on FPGA, offloading cryptography from a PicoRV32 RISC-V core to achieve **100% CPU freedom** during encryption execution.
- Engineered an **8-lane SPI interface** delivering 8× throughput over standard SPI (50 Mb/s vs 6.25 Mb/s) and a custom MMIO bridge enabling CPU-coprocessor communication via PCPI custom instructions.
- Integrated **Diffie-Hellman key exchange** for dynamic session key negotiation at runtime, eliminating hardcoded key vulnerabilities; benchmarked against SiFive E31, lowRISC Ibex, and OpenHW CV32E40P cores.
- **My Role:** FPGA interfaces 8-lane SPI slave, SPI-to-CPU MMIO bridge, RX buffer, and PCPI coprocessor interface in Verilog.

Projects

Multi-Cycle Pipeline Processor | Verilog

- Designed and implemented a multi-cycle pipelined processor in Verilog with fetch, decode, execute, memory, and write-back stages, handling data hazards via forwarding and control hazards via branch prediction.

Single-Cycle Processor with Full Hazard Handling | Verilog

- Built a single-cycle processor in Verilog implementing complete hazard resolution — data hazards via forwarding unit and stall logic, structural hazards via resource separation, and control hazards via flush on branch/jump.

Workify | React, TypeScript, Node.js, MongoDB, Redis

Live

- System Design Project: Architected a scalable freelancing marketplace with focus on availability, caching, indexing
- Designed APIs, JWT-based authentication, and optimized data models with Redis caching for efficient read/write operations.

Social Media Application | MERN Stack, Socket.io, GridFS

- Developed platform with posts, likes, comments, reels, and stories. Implemented JWT auth and GridFS for media storage.

Real-Time Chat Application | MERN Stack, Socket.io

- Built instant messaging with real-time communication, online/offline status tracking, and media sharing.

AI Document Assistant "MERNBot" | MERN Stack, OpenAI API

- Integrated OpenAI GPT-3.5 for document summarization and AI-powered Q&A with RESTful APIs.

Autonomous Vehicle System | *Raspberry Pi, OpenCV, Python*

- Designed self-driving car prototype using classical CV for obstacle recognition, edge detection, and lane-following.

Mission Logging System | *Flutter, Django, SQL*

- Developed cross-platform mobile app for Pakistan Army with Django backend.

Audio Classification CNN | *TensorFlow, Keras*

- Built deep learning model for sound classification using UrbanSound8K dataset with 87% accuracy.

Technical Skills

Hardware & Embedded: Verilog, Assembly language, FPGA (Vivado), RISC-V, Raspberry Pi, Arduino

Web & Full-Stack: MERN Stack, RESTful APIs, JWT, Redux Toolkit, TypeScript, Socket.io, Tailwind CSS

Backend: Node.js, Express.js, Django, Flask, Multer, bcrypt

Databases: MongoDB, MySQL, PostgreSQL, Redis, GridFS

Cloud & DevOps: Docker, GitHub Actions, Vercel, Render, Azure Blob Storage

Machine Learning: TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas, Matplotlib

Mobile: Flutter, Dart

Programming: Python, JavaScript, TypeScript, C++, Java

Tools: Git, GitHub, MLflow, DVC, Raspberry Pi, Arduino, OpenAI API

Certifications & Leadership

- **Director, Web & IT Wing** – COMPPEC (NUST Computer Engineering Society), Jan 2025 – May 2025
- **Machine Learning Specialization** – Coursera (Andrew Ng)
- **Deep Learning Specialization** – Coursera (Andrew Ng)
- **AWS Cloud Technical Essentials** – Coursera (AWS)